

ME 6102- Project Stage IV

Design and Development of Self-Balancing Mobile Robot

Under the Guidance of
(Prof. Prasanna S Gandhi)

Presented by- Group 1

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OUTLINE

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Motivation

- ✱ *To study the real world application of inverted single pendulum system*
- ✱ *Demonstrates practical implementation of PID control, sensor fusion, and real-time embedded processing.*

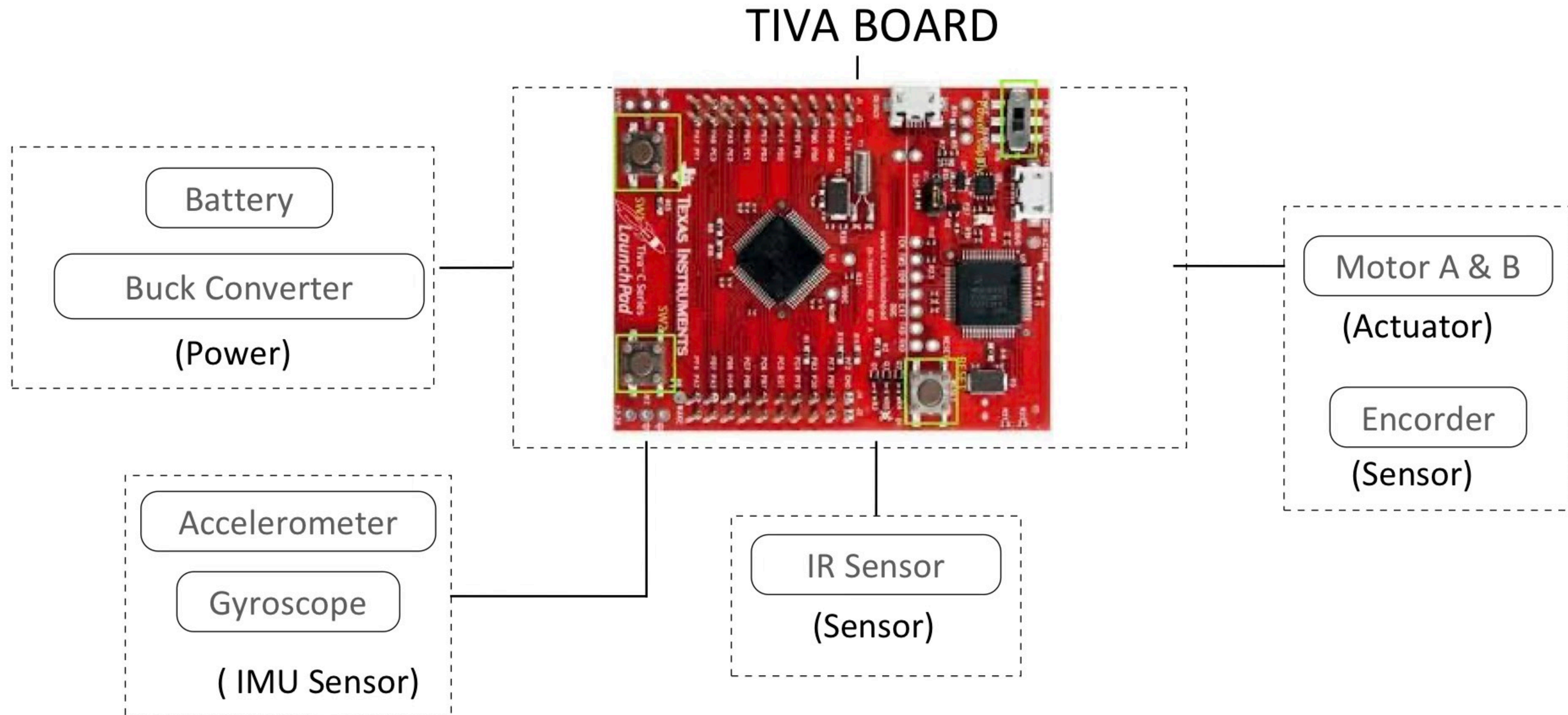
Objective

Design and development of a robot with TIVA microcontroller capable of self balancing and line following capabilities

Methodology

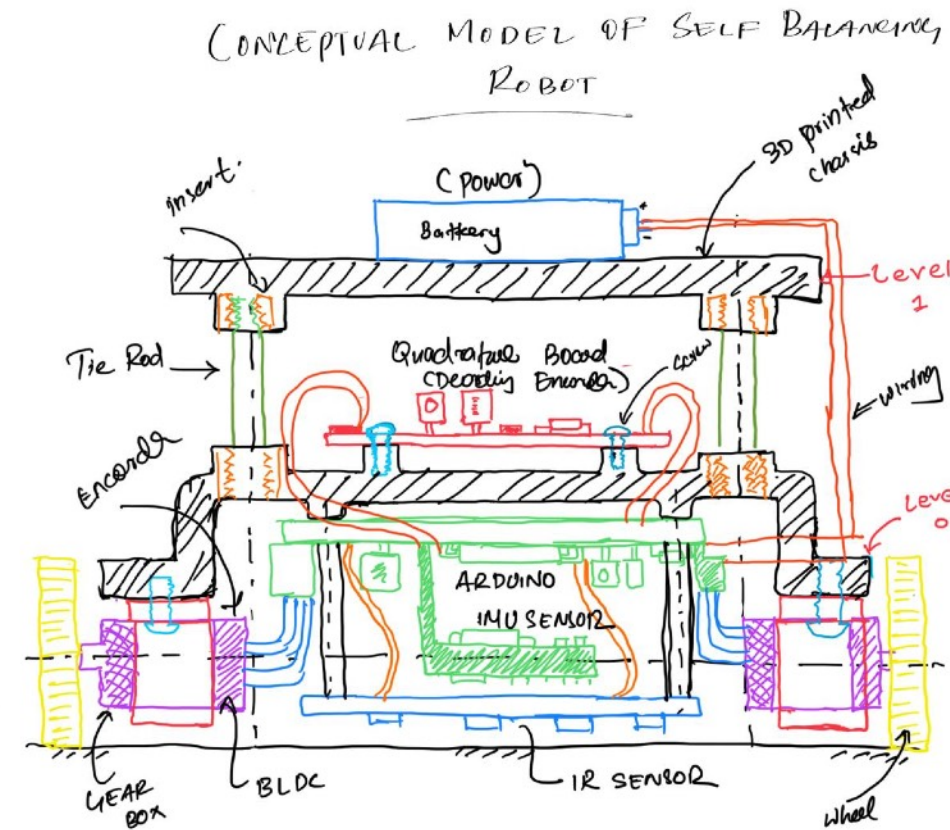
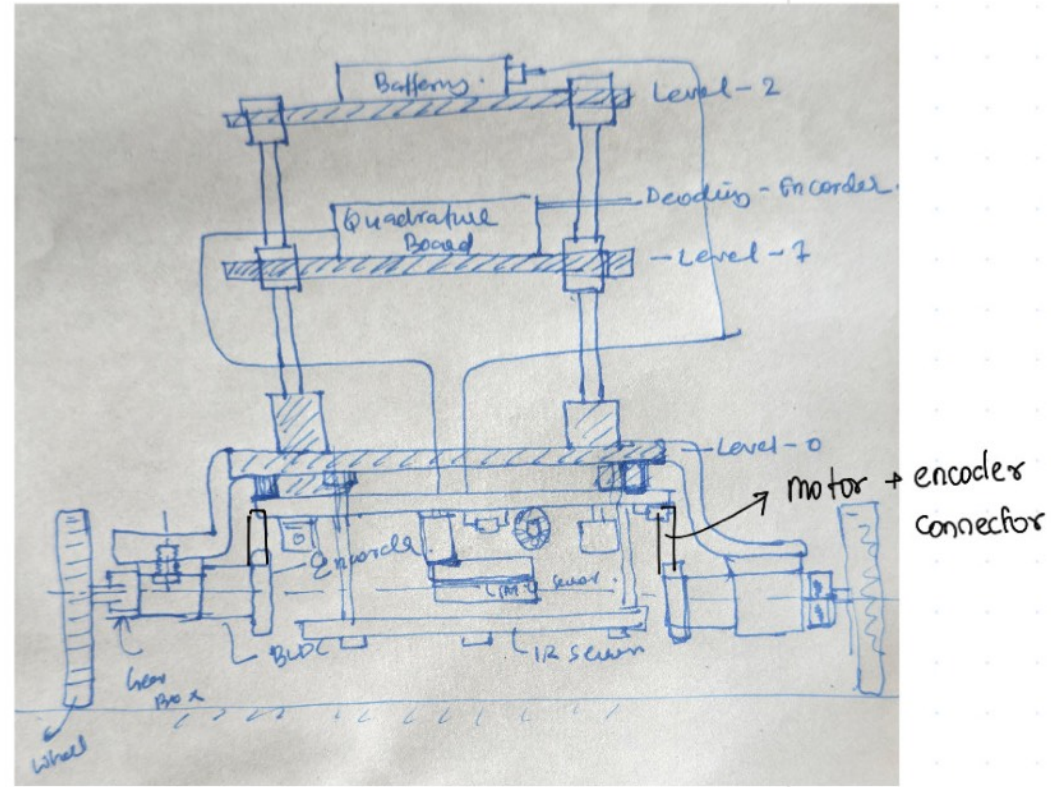
- ✱ *We used the Tiva C Series microcontroller as the main controller for all real-time operations.*
- ✱ *The robot integrates IMU for tilt estimation and encoders for wheel speed feedback.*
- ✱ *The mechanical structure is made using laser-cut acrylic plates assembled with spacers.*
- ✱ *The electrical system (motor driver, buck converter, sensors) is neatly integrated with the chassis.*
- ✱ *The firmware was developed in Code Composer Studio using TivaWare driver libraries.*

System Layout



Conceptualisation

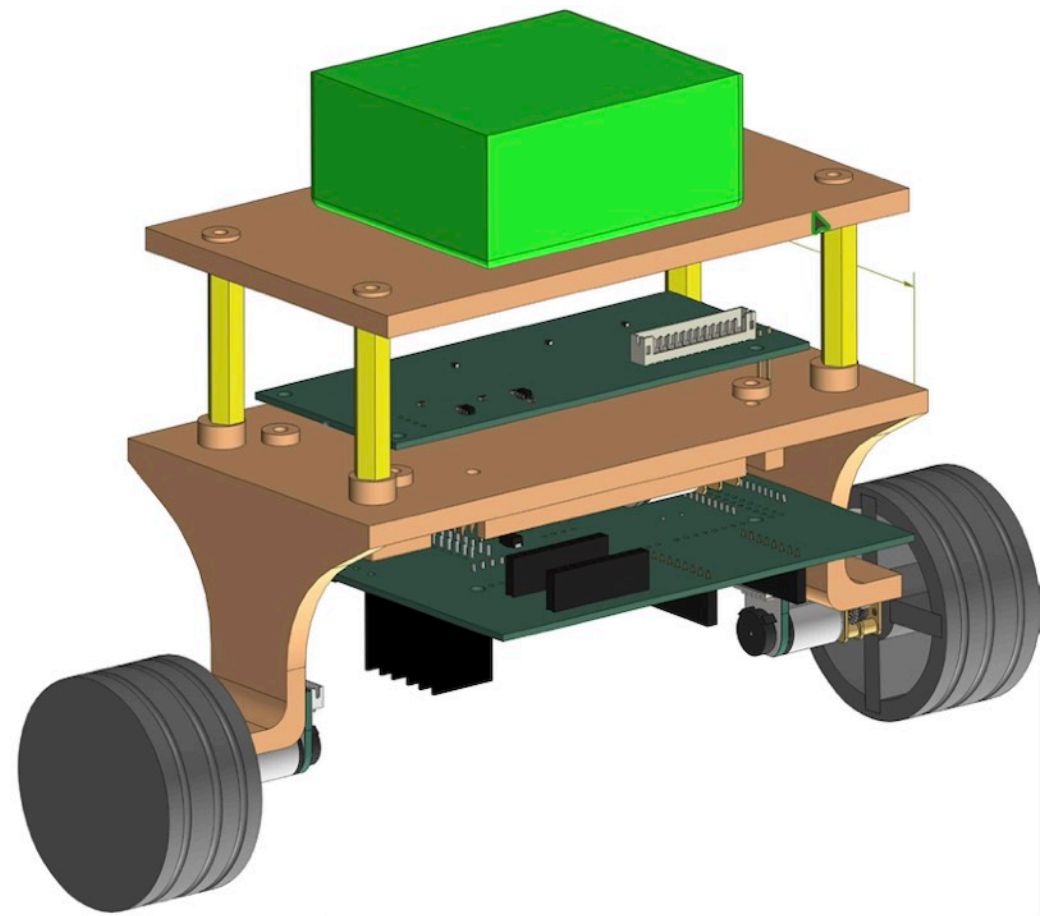
Version 1



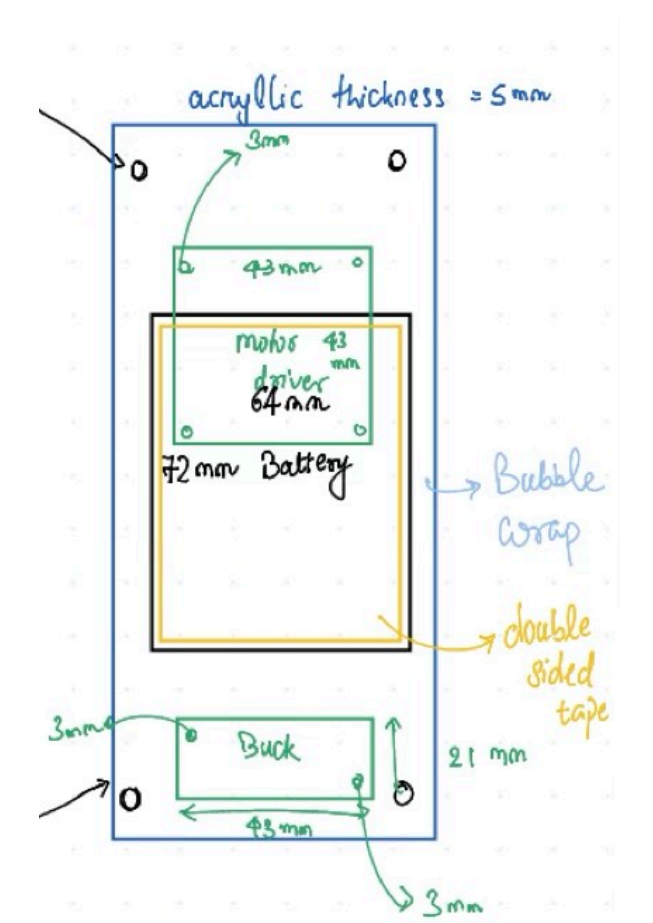
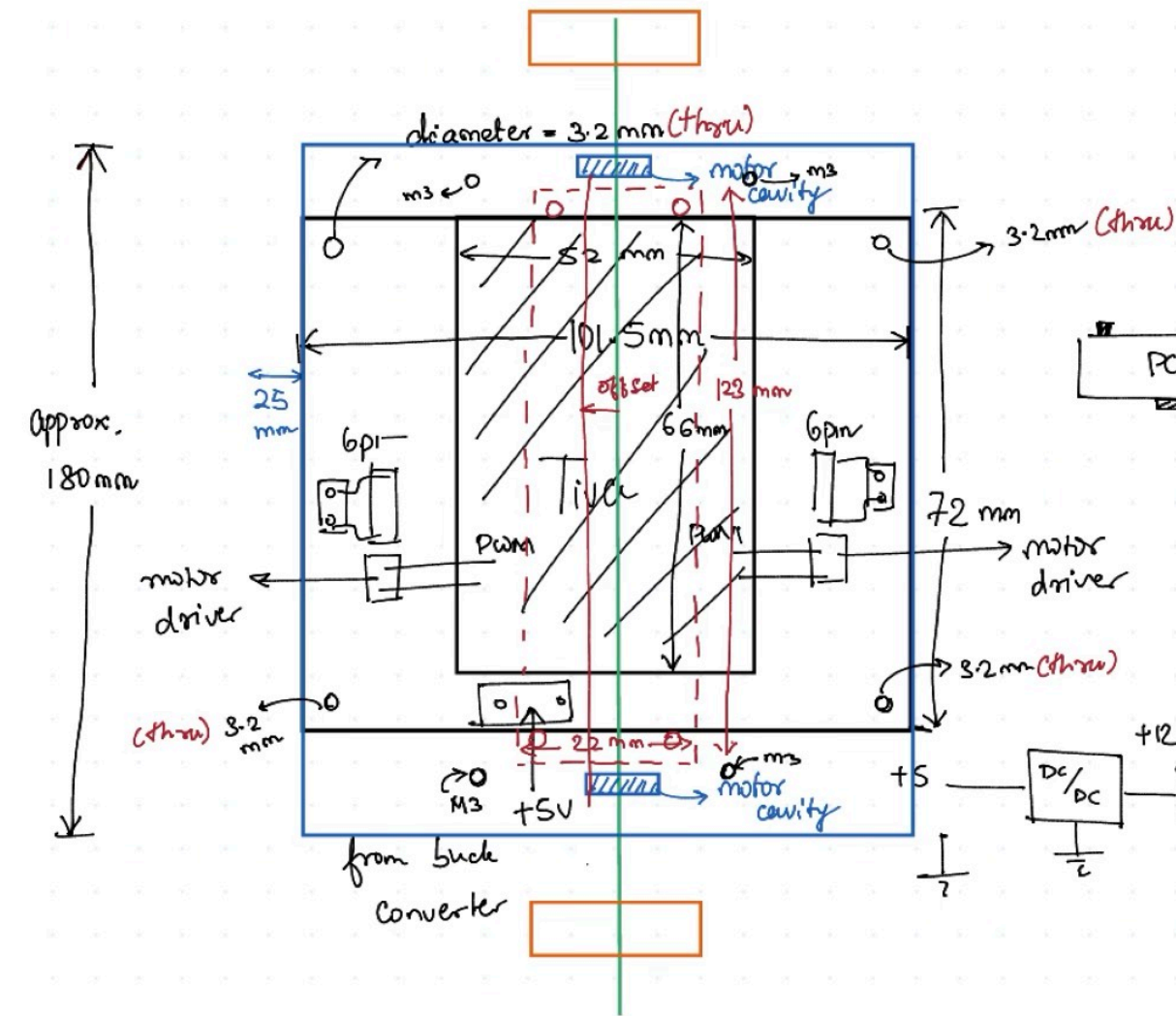
Fabrication : 3D printing
Level : 2
Microcontroller : Arduino

Challenges

- ## * Arduino processing power limitation



Version 2



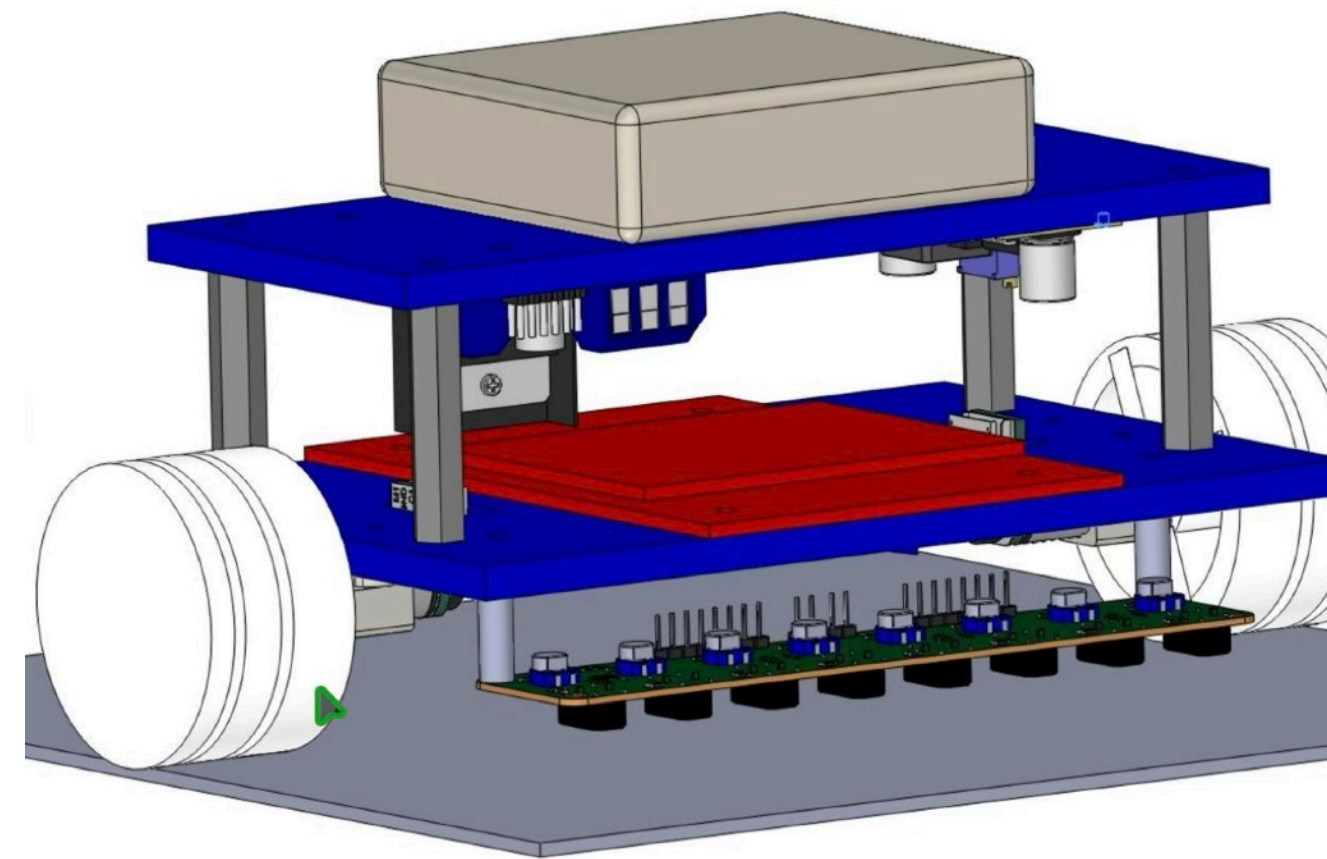
Fabrication : Acrylic

Level : 1

Microcontroller : TIVA

Advantages

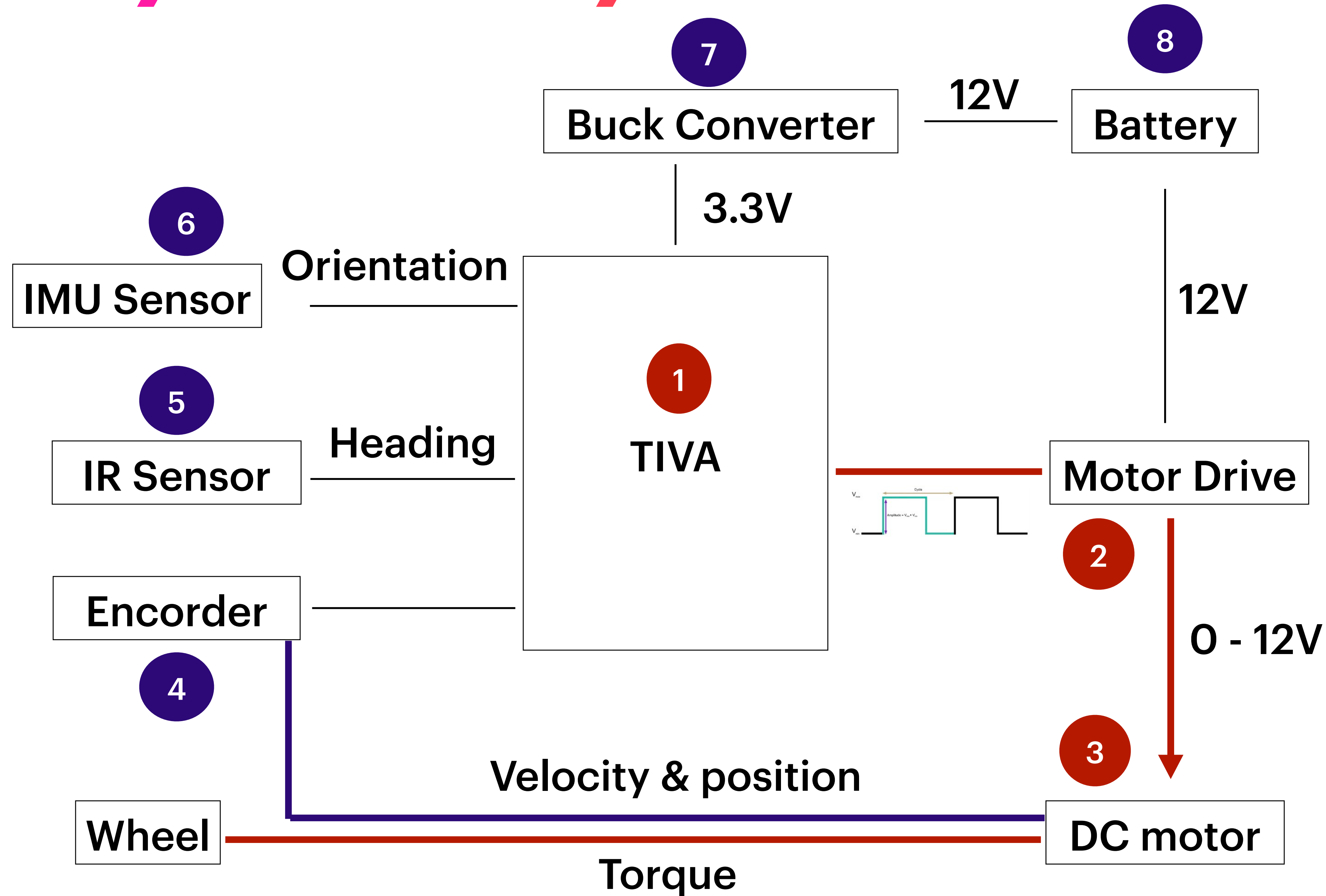
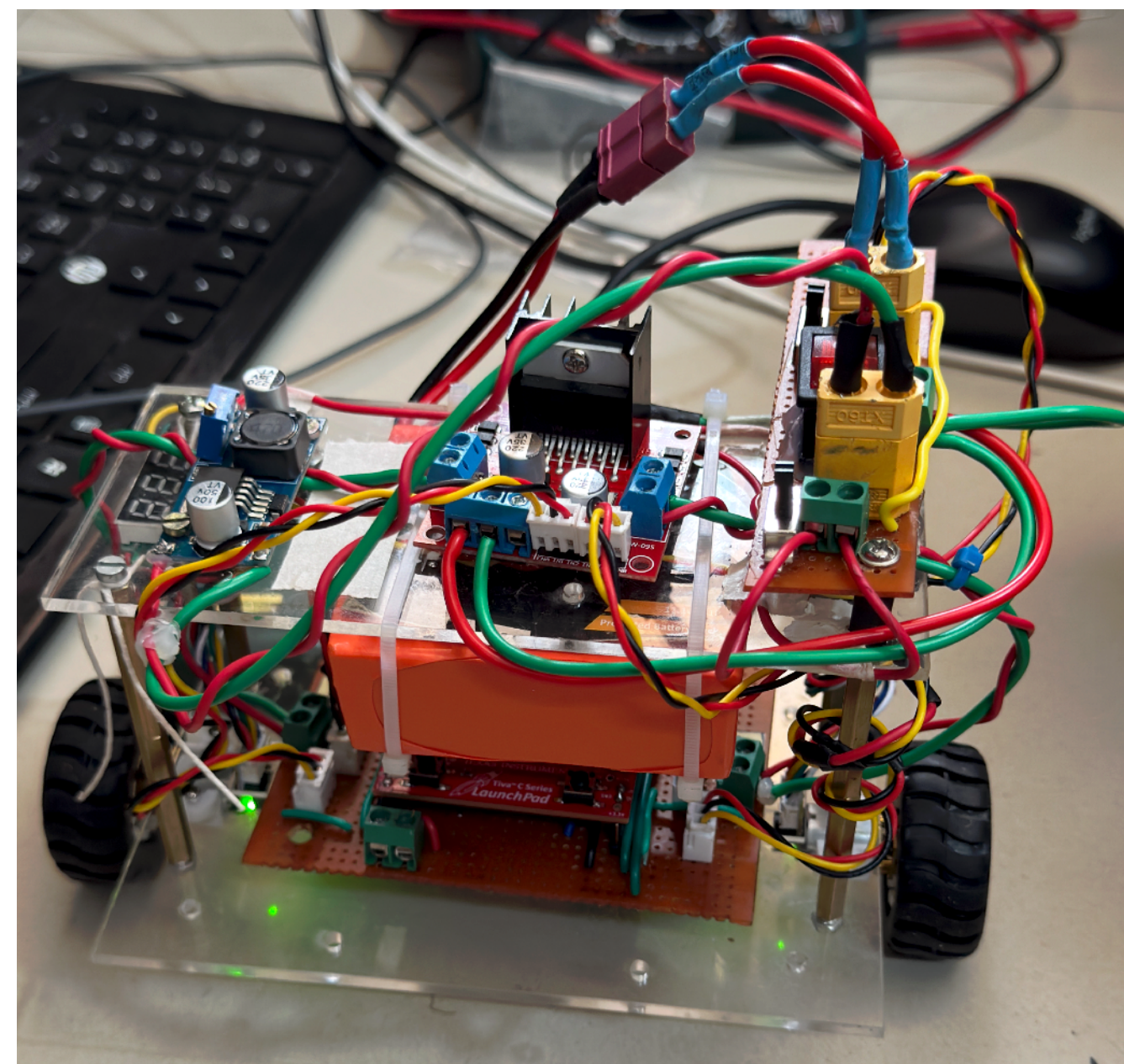
- ✱ Simple fabrication
- ✱ Hardware availability
- ✱ Fast Realisation



Specifications

- ✱ Microcontroller : TIVA C series
- ✱ Motor : GA12- N20 DC Motor (12V)
- ✱ Buck Converter - LM2596S
- ✱ Motor Drive - L298N
- ✱ Wheel - N20 Wheel 44mm dia
- ✱ Sensors
 - Encoder : Pro-range OE-28
 - IMU - MPU 9250
 - IR - RLS 08 Analog & Digital Line sensor array

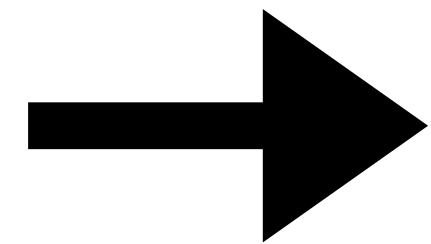
System Layout



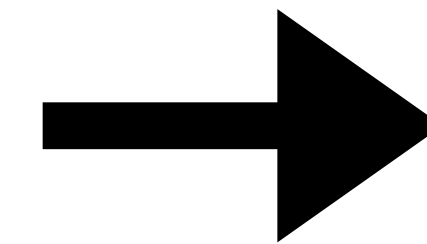
Fabrications

Mechanical integration

**3D CAD
Modelling**



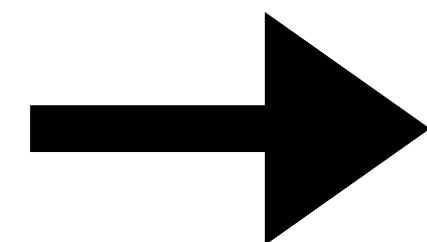
DXF File



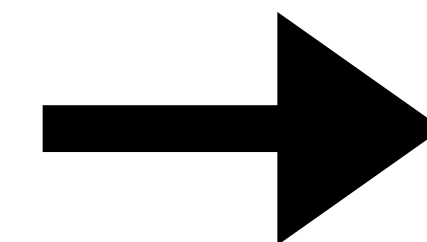
**CO2 Laser
Cutting**

Electrical integration

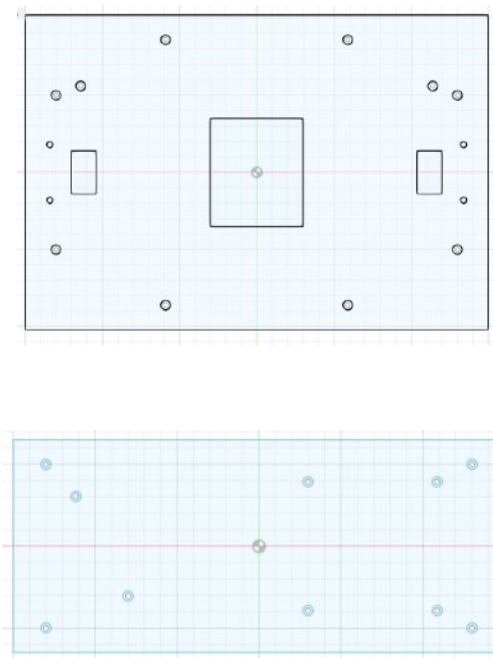
**Electrical
components**



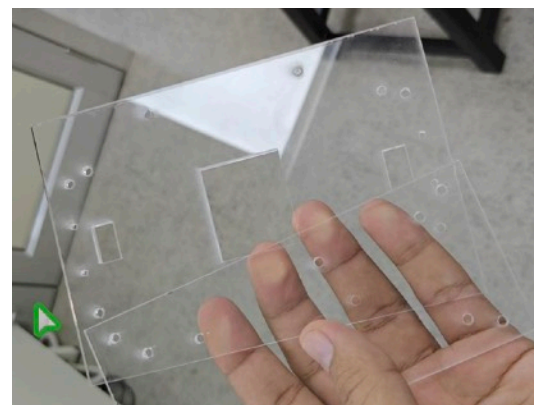
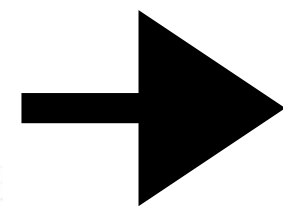
Soldering



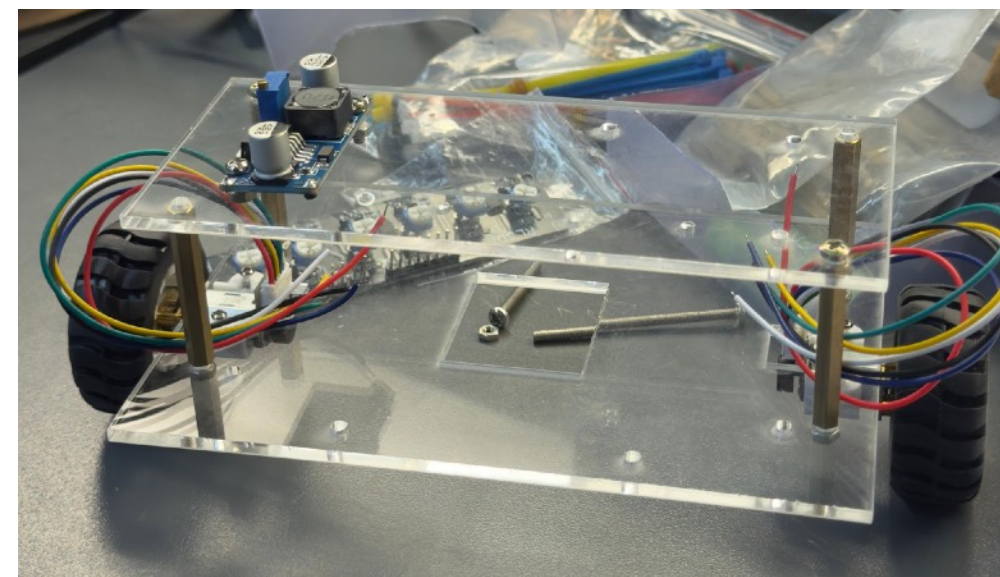
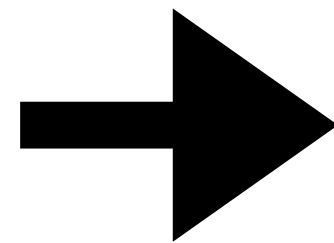
PCB + wiring



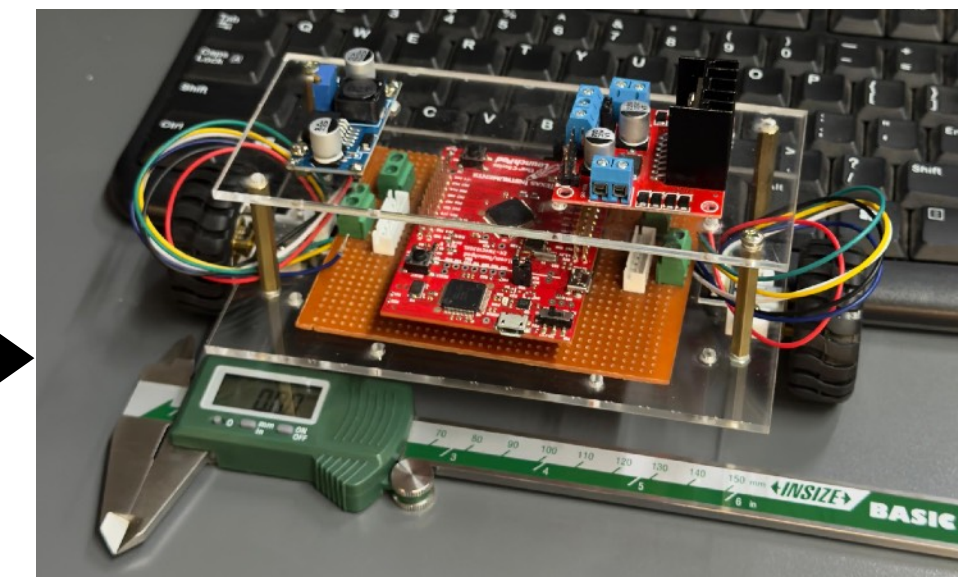
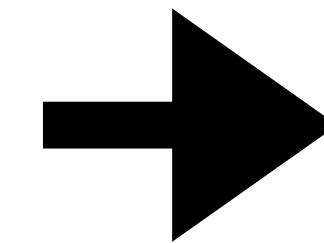
3D CAD



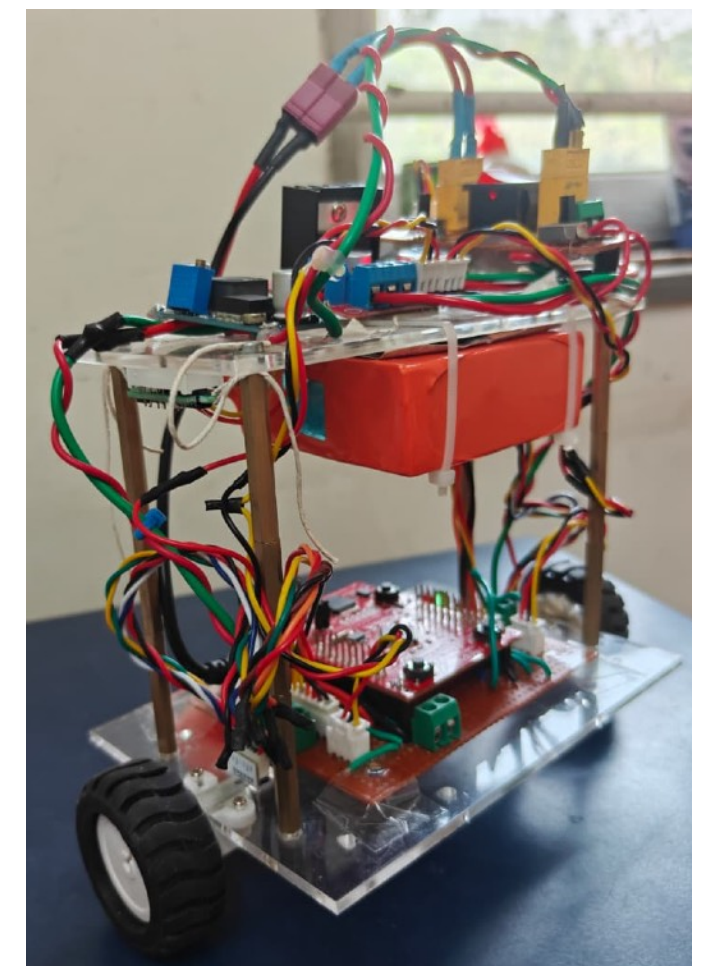
Laser cut acrylic
sheet



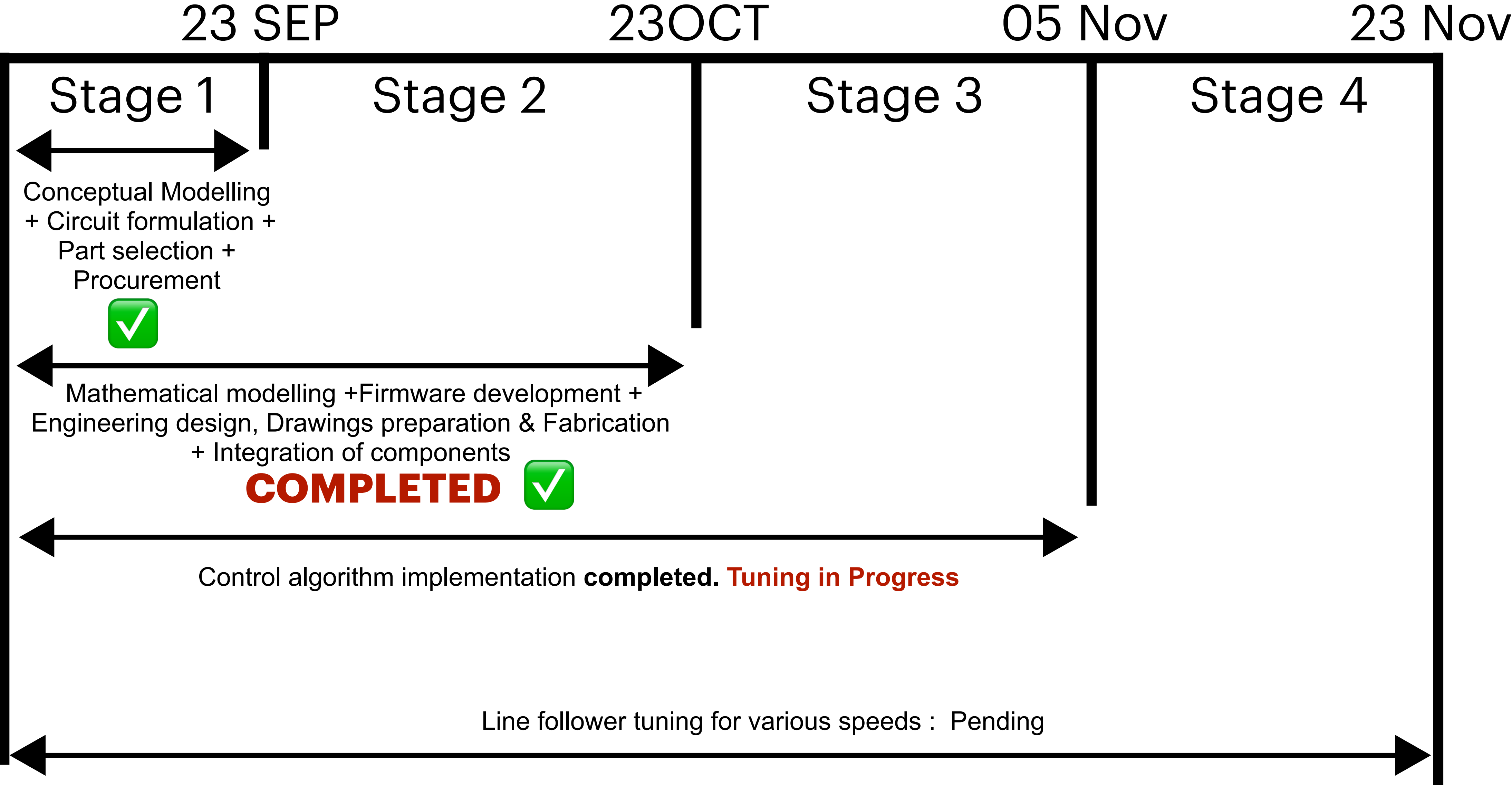
Assembly of both the
levels



Assembly of TiVA
Microcontroller

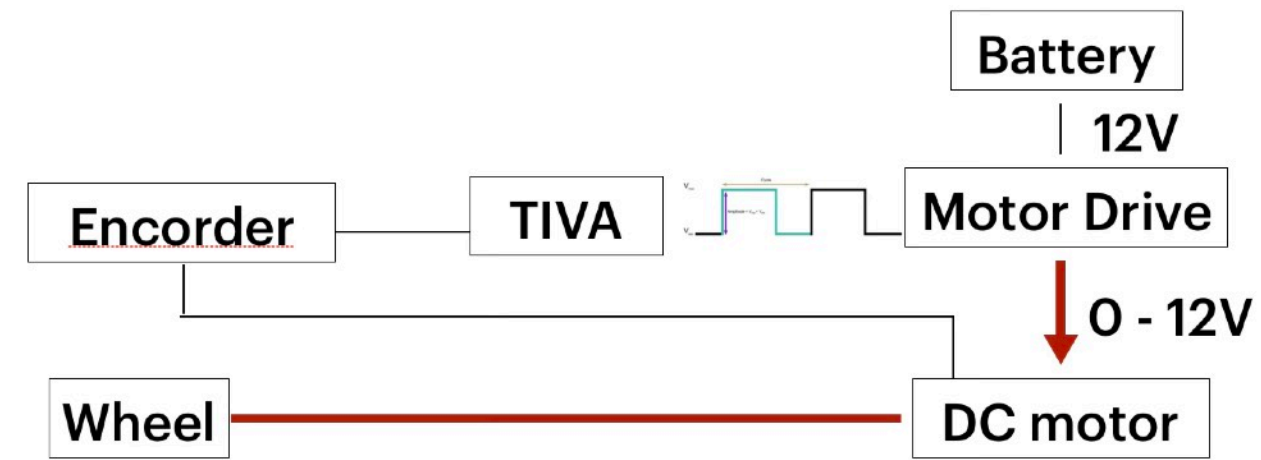


PROJECT PIPELINE



Wheel Control

Software



Hardware

Micro Controller

- * TIVA C- Series

Battery

- * 12V - Lithium ion battery
- * 2500 mAH

Motor drive

- * L 298N
- * Voltage O/P: 0 - 12 V
- * Gear Ratio - 250:1

Motor

- * N20 DC motor
- * 12 V, 0.6A
- * Torque = 2N-cm

Encorder

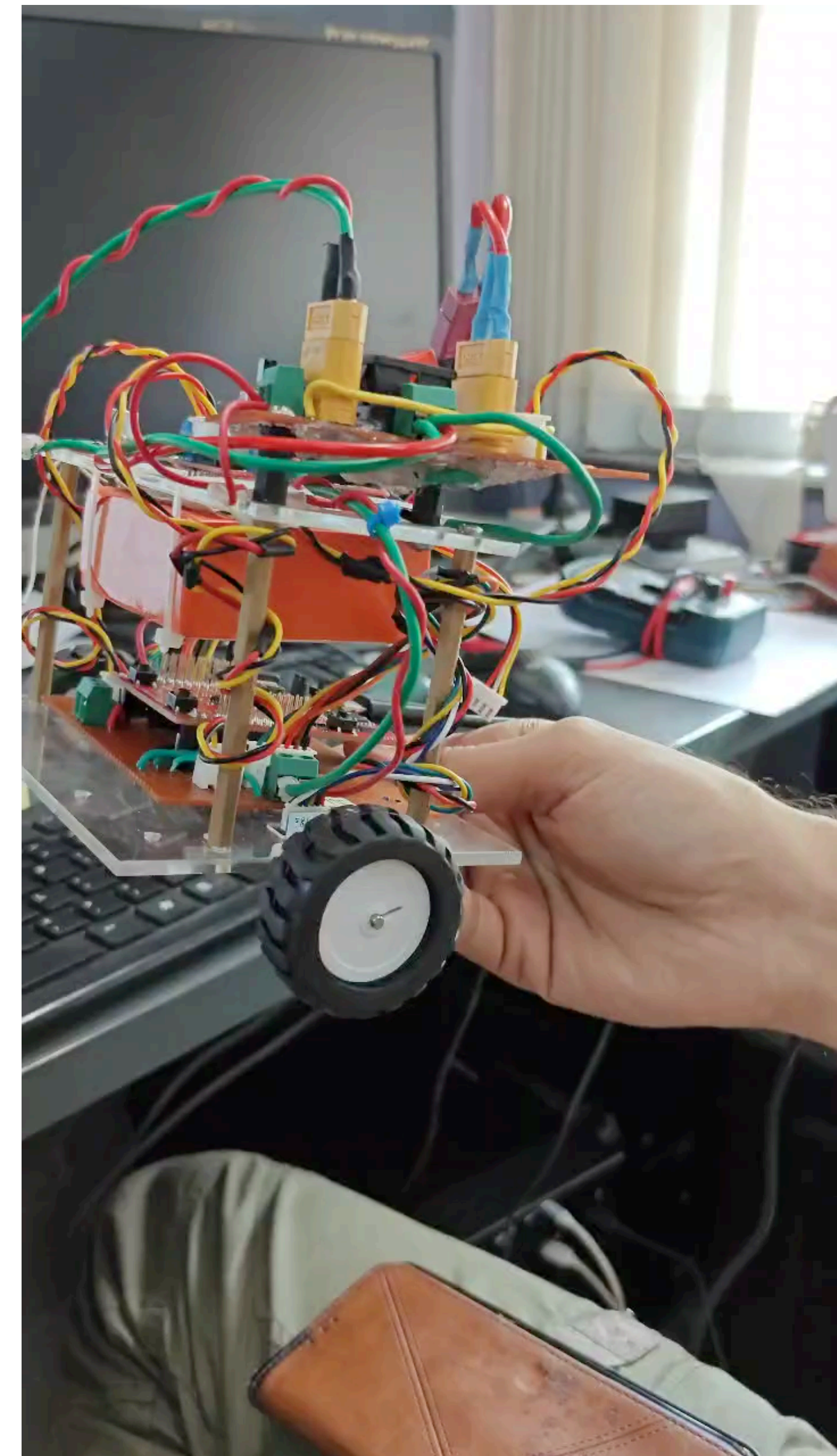
- * 3 Pulse Per Revelation.

Software : Code Composer Studio

Objective : Intuitive wheel control

Current Status : PWM generation and Encorder reading completed

➡ **Wheel control code**



BALANCE CONTROL

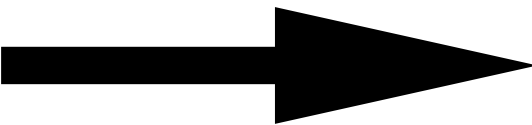
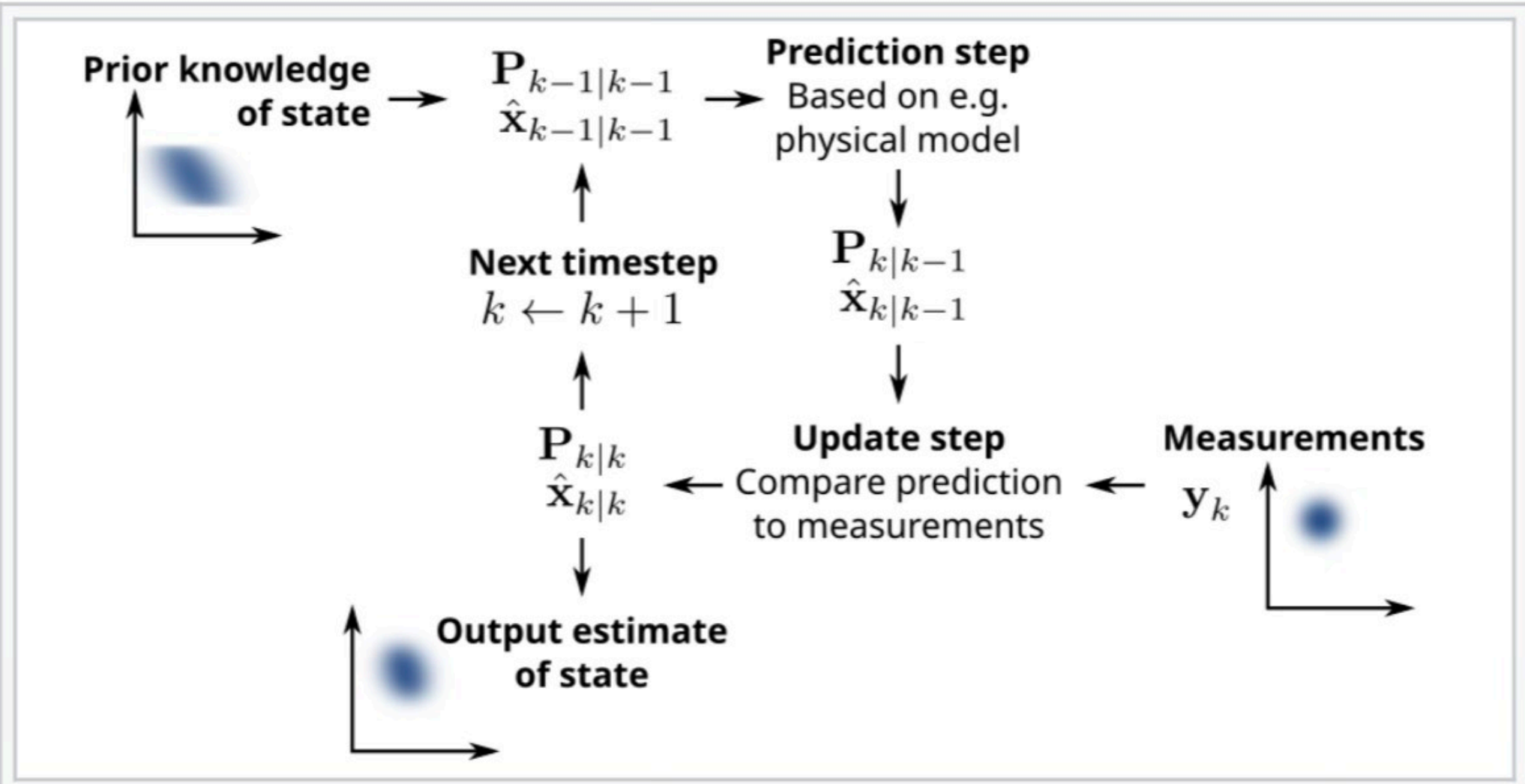
Hardware

✱ Inertial Measurement Unit

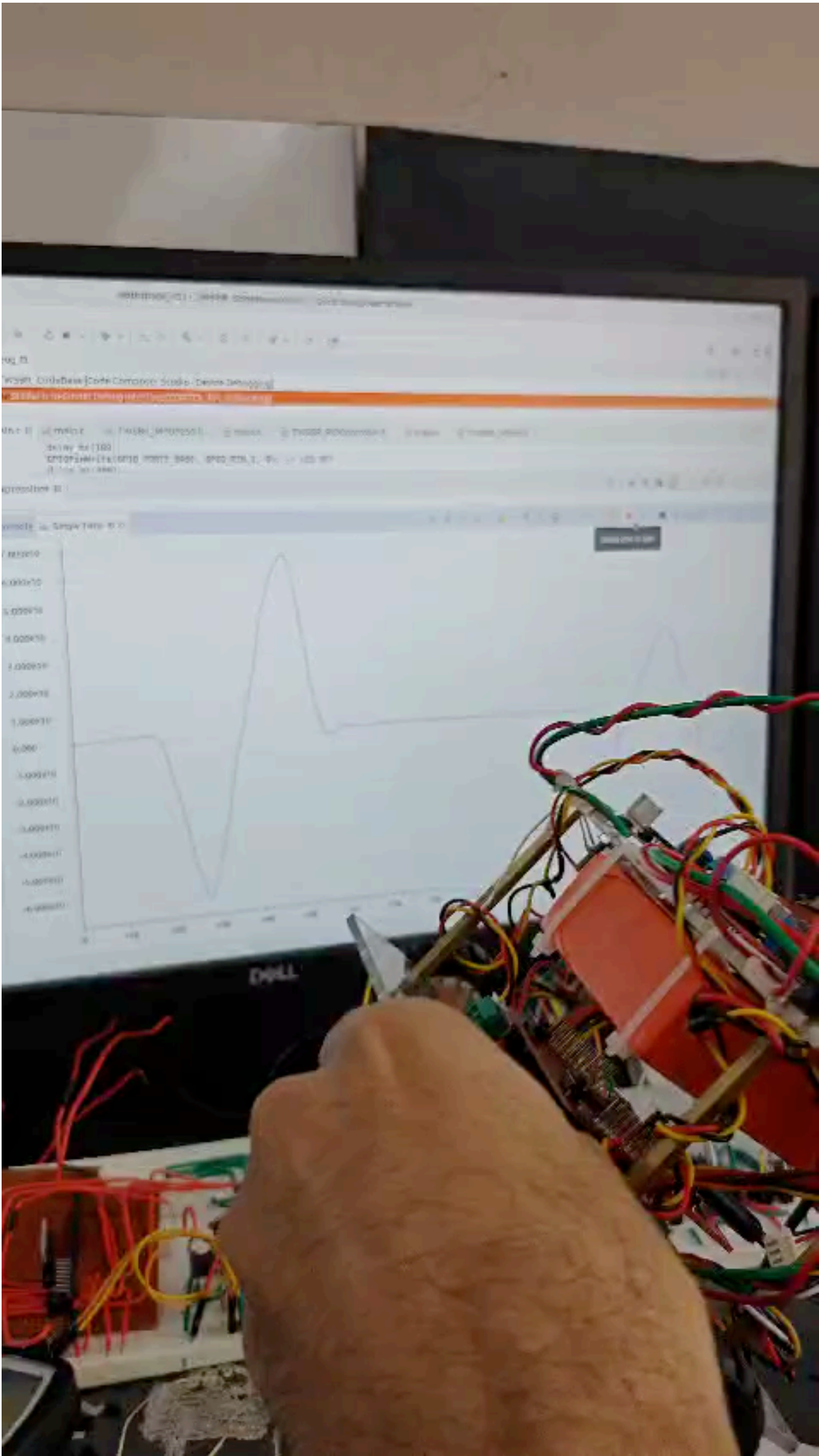
Software

Objective : To obtain the accelerometer and gyroscope reading and to use them to estimate the angle of tilt.

Algorithm : Kalman filter



Kalman filter code



Mathematical modelling - Dynamics

NON-LINEARISED DYNAMIC EQUATIONS

$$\ddot{x} = - \frac{m^2 g l^2 \sin \alpha \cos \alpha}{(M+m+2(M_w + \frac{I_w}{r^2}))(ml^2 + I_p) - m^2 l^2 \cos^2 \alpha} + \frac{m \dot{\alpha}^2 l \sin \alpha (ml^2 + I_p)}{(M+m+2(M_w + \frac{I_w}{r^2}))(ml^2 + I_p) - m^2 l^2 \cos^2 \alpha} + \frac{\tau_L + \tau_R}{r} \cdot \frac{ml^2 + I_p + ml r \cos \alpha}{(M+m+2(M_w + \frac{I_w}{r^2}))(ml^2 + I_p) - m^2 l^2 \cos^2 \alpha}$$

$$\ddot{\alpha} = \frac{(M+m+2(M_w + \frac{I_w}{r^2})) m g \sin \alpha l}{(M+m+2(M_w + \frac{I_w}{r^2}))(ml^2 + I_p) - m^2 l^2 \cos^2 \alpha} - \frac{m^2 \dot{\alpha}^2 l^2 \sin \alpha (ml^2 + I_p) \cos \alpha}{(M+m+2(M_w + \frac{I_w}{r^2}))(ml^2 + I_p) - m^2 l^2 \cos^2 \alpha} - \frac{ml \cos \alpha + (M+m+2(M_w + \frac{I_w}{r^2})) r}{(M+m+2(M_w + \frac{I_w}{r^2}))(ml^2 + I_p) - m^2 l^2 \cos^2 \alpha} (\tau_R + \tau_L)$$

LINEARISED DYNAMIC EQUATIONS (LQR)

$$\Delta \ddot{x} = - \frac{m^2 g l^2}{\left[(M+m+2(M_w + \frac{I_w}{r^2}))(ml^2 + I_p) - m^2 l^2 \right]^2} \Delta \alpha + \frac{(ml^2 + I_p + ml r)}{\left[(M+m+2(M_w + \frac{I_w}{r^2}))(ml^2 + I_p) - m^2 l^2 \right]^2} \frac{\Delta \tau_L + \Delta \tau_R}{r}$$

$$\Delta \ddot{\alpha} = \frac{(M+m+2(M_w + \frac{I_w}{r^2})) m g l}{\left((M+m+2(M_w + \frac{I_w}{r^2}))(ml^2 + I_p) - m^2 l^2 \right)} \Delta \alpha - \frac{ml + (M+m+2(M_w + \frac{I_w}{r^2})) r}{\left((M+m+2(M_w + \frac{I_w}{r^2}))(ml^2 + I_p) - m^2 l^2 \right)} (\Delta \tau_L + \Delta \tau_R)$$

Mathematical modelling - Kalman Filter

$x \rightarrow$ tilt angle , $w \rightarrow$ tilt rate

$$P_{0/0} = 1, \quad x_{0/0} = 0$$

$$\hat{x}_{k|k-1} = A \hat{x}_{k-1|k-1} + B u_k, \quad A = 1, \quad B = \Delta t, \quad u_k = w$$

$$P_{k|k-1} = A P_{k-1|k-1} A^T + Q, \quad Q = \text{process covariance}$$

$$K_k = P H^T (H P H^T + R)^{-1}, \quad H = 1, \quad K_k: \text{Kalman gain}$$

, $R = \text{measurement error covariance}$

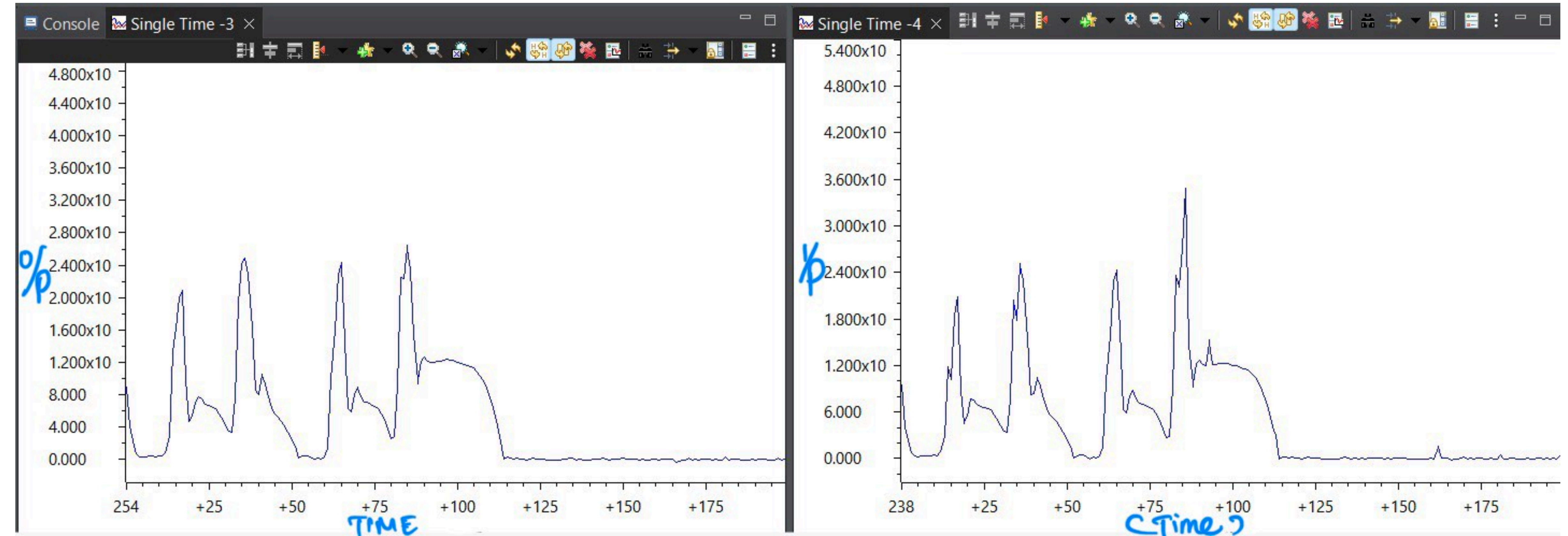
$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k (z_k - H \hat{x}_{k|k-1})$$

, $z_k = \text{pitch from accelerometer}$

$$= \tan^{-1} \left(\frac{a_y}{a_z} \right)$$

$$P_{k|k} = (I - K_k H) P_{k|k-1}$$

Input and output of kalman filter



UNMET OBJECTIVES

- ✱ **Line-Following Feature Not Integrated**
- ✱ **Complete Stability Not Achieved**

Reason for not meeting the objectives

- ✱ *Despite integrating the IMU, motor driver, and Tiva C controller successfully, the robot did not reach complete stability because the PID control parameters (K_p , K_i , K_d) could not be fully optimized.*
- ✱ *We performed multiple experiments by varying these gains, but the system continued to show oscillations and drift.*
- ✱ *Achieving stable balance requires extensive tuning under different loading and surface conditions, which we could not complete within the available time.*
- ✱ *Therefore, additional testing and further PID refinement are needed to meet the stability objective*

Reference

1. A modified dynamical formulation for two-wheeled self-balancing robots --Ali Ghaffari · Azadeh Shariati · Amir H. Shamekhi.
2. Dynamic Analysis of a Nonholonomic Two-Wheeled Inverted Pendulum Robot-- Yeonhoon Kim, Soo Hyun Kim and Yoon Keun Kwak.

Thank you